Moustafa Mohammed Abdelwhab

B.Sc. Computer Science — Arab Academy for Science, Technology & Maritime Transport, Cairo
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PROFESSIONAL SUMMARY

Software engineer who designs and delivers complete systems — web front ends, backend services and REST APIs, relational and NoSQL data layers, and containerized deployment. Applied background in data mining and machine learning: classification, clustering, forecasting, and anomaly detection. Seeking a software engineering role .

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, C++, Java, SQL

Frontend: React, Next.js, HTML5, CSS3, Tailwind, Bootstrap

Backend: Node.js, Express, RESTful API design, MQTT

Databases: PostgreSQL, MySQL, MongoDB, InfluxDB (time-series), Supabase

Data & ML: Data mining, classification, clustering, forecasting, anomaly detection, NLP embeddings

Tools & Cloud: Docker, Docker Compose, Git, AWS, Vercel

Networking: CCNA coursework — routing, switching, network fundamentals

PROJECTS

Smart Plug Energy Monitoring & Management System — Graduation Project 2025 – 2026

Node.js · Express · Next.js · MQTT · InfluxDB · PostgreSQL · MongoDB · Docker · AES

- Built an end-to-end IoT platform where smart plugs publish per-second readings over MQTT to a backend serving a live monitoring dashboard, deployed as an eight-service Docker Compose stack.
- Designed a three-database architecture: InfluxDB for time-series readings, PostgreSQL for relational records, MongoDB for the aggregated history powering analytics. Secured transport with AES encryption.
- Developed ML models for consumption forecasting, anomaly detection, and savings recommendations, plus a device simulator enabling full-system testing without physical hardware.
- Team of six — led the architecture and implemented the backend, analytics pipeline, ML models, and deployment.

Wedding Photos — Guest Photo-Sharing Web Application 2026

React · Vite · Supabase · Vercel

- Built and deployed a single-page application letting event guests upload photographs directly to shared cloud storage without an account, using Supabase bucket policies and continuous deployment on Vercel.
- Live: [https://lamaj-wedding-photos.vercel.app/]

Competition Hosting Platform — SaaS Product 2024

React · Node.js · PostgreSQL

- Building a multi-tenant platform for organizing competitions end to end: player and team registration, bracket and fixture scheduling, match result tracking, and role-based dashboards for organizers.

Landing Pages & Marketing Front Ends 2023 – Present

React · Next.js · Tailwind CSS · Framer Motion · GSAP · Vercel

- Design and build responsive, conversion-focused landing pages — hero and pricing sections, scroll-triggered animations, and lead-capture forms — using Tailwind, shadcn/ui, Framer Motion, and GSAP, with attention to Core Web Vitals and SEO.

COURSES & CERTIFICATIONS

Front-End Web Development — React · Bootstrap · JavaScript · HTML5 & CSS3
CCNA — Networking Coursework · routing, switching, and network fundamentals
Cloud Computing — AWS services: EC2, S3, Lambda, IAM, VPC, RDS, CloudWatch

EDUCATION

B.Sc. Computer Science — Arab Academy for Science, Technology & Maritime Transport Oct 2022 – 2026

Cairo, Egypt. Graduation Project: Smart Plug Energy Monitoring & Management System.

Languages: Arabic — Native | English — Professional working proficiency